Vellore Institute of Technology
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School of Computer Science Engineering and Information Systems

(SCORE)

INFORMATION SECURITY ANALYSIS AND AUDIT – CSE 3501

PROJECT TITLE: NAS BASED RANSOMEWARE DETECTION AND MONITORING SYSTEM

EPJ Component – Final Review

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ABSTRACT:

Securing the system focuses on protecting sensitive data from unauthorized access, tampering, or exploitation by external and internal threats. The main goal is to safeguard critical information and to stop ransomware attacks that can harm data integrity and availability.

Ransomware is a type of malicious software that encrypts or locks users' files, making them inaccessible until a ransom is paid to the attacker. Such attacks can lead to significant data loss, financial harm, and service disruptions. To address this, we have created a system that is designed to monitor files and system behavior in real-time. It looks for unusual or harmful activities that might signal ransomware.

By using machine learning algorithms, the system can analyze file access patterns, detect suspicious changes, and determine if an operation is legitimate or potentially harmful. When it spots ransomware-like behavior, the system will quickly isolate the affected files or devices to stop the malware. It will also send alerts to the dashboard and via email.

Along with the software detection layer, this system includes IoT components, such as esp32 microcontroller, NAS, sd module reader, LED'S to ensure a practical and automated response.

Overall, this approach offers a multi-layered defense that combines data protection, machine learning-based anomaly detection. It helps ensure that the system stays strong against ransomware and other cyber threats while keeping operations efficient and maintaining user trust.

INTRODUCTION:

In today's digital world, safely storing and accessing data is very important. Network Attached Storage (NAS) is a system that helps users store and share files over a network. It is commonly used in homes, offices, and organizations to keep data in one place. Cybercriminals are increasingly using ransomware attacks to gain access to a victim's data, encrypt it, and demand payment [1]. This is crucial for organizations since it can lead to financial loss. Often, they cannot access the stored data because of additional encryption placed by hackers.

To protect NAS from such attacks, we can use a ransomware detection system that works with IoT (Internet of Things) devices. By integrating with IoT devices, this system scans the files for ransomware. If it finds any, it immediately alerts the user. If no ransomware is detected, the files are stored on an external NAS with the user's permission. The system continues to monitor the files using file monitor

algorithms, checking for changes in entropy and hashing. The Network Attached Storage is typically connected in a server format, with the ESP32 and the main uploading and monitoring system linked to the same local host.

The system is designed to monitor file behavior, identify suspicious patterns, and alert users immediately through the mail and dashboard. The storage module, along with external storage, acts as NAS where files are stored securely. After checking and storing the files with the help of monitoring algorithms, the system continues to monitor the saved files.

MOTIVATION:

The NAS based ransomeware detection system has been majorly developed to store their data in secure manner with external storage to monitor and detect ransomeware. So this was the main reason to create the ransomeware detection system. Ransomware attacks targeting NAS devices have several ways to exploit the vulnerabilities in the system due to shared access and network connectivity. Some of the major reasons to create the system for monitoring and detection of ransomeware includes the following.

- Weak and outdated passwords make the system easy to get exploited to threats.
- Overall there are only the minor ransomeware based protection system not focused on monitoring.
- Most of the security related tools often fail to detect them when normal scan of the storage systems.
- This will result in the serious financial loss for the organizations.
- Absence of continuous monitoring of the files lead to inaccessible data.

The goal of this project is to reduce the threat of ransomware attacks on NAS systems. Most NAS systems do not provide real-time protection. As a result, they are vulnerable to file encryption. Once they are infected, data can become completely inaccessible. This can lead to financial loss, service downtime, and permanent data damage.

This system continuously watches file behavior, checks files before they are saved, and sends alerts when it detects unusual activities like mass file renaming or encryption. It uses microcontrollers like the ESP32, a storage module, and buzzers to provide immediate feedback. Additionally, this system requires secure user authentication before allowing uploads. This ensures that only authorized users can store files.

OBJECTIVE:

The main goal of this project is to design and implement a ransomware detection and monitoring system for Network Attached Storage using IoT devices and cybersecurity methods. This system protects sensitive files on NAS from unauthorized access, suspicious file activity, and possible ransomware attacks.

- To monitor NAS file storage for unusual behavior, such as continuous renaming or encryption.

- Restrict access to the storage system to authorized users only.

- Send real-time logs and alerts to the users before and after file saving.

- To create a cost-effective ransomware detection and monitoring system [4].

- To detect and block ransomware attacked files before they are saved or accessed.

- Using machine learning algorithms to improve accuracy of the ransomeware detection system [17].

The major objective of the NAS ransomeware detection system focuses on the above to reduce data and financial losses.

EXISTING SYSTEMS:

Detecting ransomeware based on disk and cpu usage data:

This system checks the cpu and storage usage of a particular system to check if the entire system or specific location is affected by ransomeware or not. This type of detection system generally requires administrative permissions to run them on our pc. After detecting ransomeware this system will isolate the specific file or folder to prevent the other files being affected and also to protect the data by providing access to only authorized users [12] .

Ransomeware detection and monitoring using ML models:

Basically the system works by using ML models to monitor the NAS storages or other system and detect the ransomeware for secure storage aspect. Mostly watchdog or inotify algorithms will be used for monitoring. Forest fire, XGboost and gradient boost etc these algorithm are trained with the json datasets determining which algorithm is best suitable with high accuary and stability, then we will select one of them to detect the ransomeware based on mass encryption, changing file names, entropy level of the files [5] .

Hybrid ML method to detect ransomeware:

Combine outputs of multiple ML models to improve detection and stability. Aggregates predictions from models analyzing CPU, disk usage, file behavior to detect ransomeware. It can reduce the false positives and can handle the various ransomeware types [5][12].

Honeypot based ransomeware detection method:

In this existing system we will creating a decoy systems that appears to be legitimate to attackers, in which they will be performing the ransomeware attacks , then they will be isolated from the original network and they will be capturing the attack methods and tools of the ransomeware. Based on this we will be analyzing the attacks to tackle them. The ransomeware attacks happen when there is continuous hashing, high level of entropy of files, when we cant able to access the data [9].

Table 1: List of ransomware hashes and families [12].

File Name	Family (MalwareBytes)	Family (Microsoft)
03bb4d5c0179fdceacc5df7645e6e7fa93e931ac9beb1968c7bbe40ba3499194	Ransom.Phobos	Ransom:Win32/Phobos.SW!MSR
0a937d4fe8aa6cb947b95841c490d73e452a3cafcd92645afc353006786aba76	Ransom.LockBit	Ransom:Win32/LockBit.PA!MTB
133bf8be0cf7003b83b03579970997d408a930e58ec2726715140520900c06de	Ransom.Sodinokibi	Ransom:Win32/Revil.SI!MTB
17d153a225ea04a229862875795eeec0adb8c3e2769ba0e05073baaf86850467	Ransom.Sodinokibi	Ransom:Win32/Revil.SI!MTB
3cff33197edc918d47d08f44d6ddbdda157337f0ad58288d15746cf72c0e4c57	Ransom.Sodinokibi	Ransom:Win32/Revil.SI!MTB
3fdad99a17a6766fe396081f82394f5e2da0142651427da64a5b6e28c9df2fd4	Ransom.Sodinokibi	Ransom:Win32/Sodinokibi.C
42c28feb23c992a350673d63413bf11bc816d00a079462ab524934219d46430d	Ransom.Sodinokibi	Ransom:Win32/Sodinokibi.DSB!MTB
4e2554b448424859b508433e6c4a043718343febc7894a849f446dd197b47d1e	Ransom.Maze	Ransom:Win32/Maze.PA!MTB
4f7bdda79e389d6660fca8e2a90a175307a7f615fa7673b10ee820d9300b5c60	Ransom.NetWalker	Ransom:Win32/Filecoder.DD!MTB
57cf4470348e3b5da0fa3152be84a81a5e2ce5d794976387be290f528fa419fd	Ransom.NetWalker	Ransom:Win32/NetWalker!MSR
5cba3e44271279e747a67dd312d4dca18832b5a850ea6b85a460846ef0101fb6	Ransom.Sodinokibi	Undetected
5d9b75e2cb84333c6b56604ce47af75b11f80bf9079054f6619251b68357d87c	Ransom.Sodinokibi	Trojan:Win32/Occamy.C
6628de7ffbbe168a4fa9ff0a1a29b54e88a32e5963db0dd1aea4b80102c8ce01	Ransom.Sodinokibi	Ransom:Win32/Revil.SI!MTB
7c2dbad516d18d2c1c21ecc5792bc232f7b34dadc1bc19e967190d79174131d1	Ransom.DeathRansom	Undetected
7d98972d5c78e1d4969da76856d6818942b606c267efa67fd31d39ae77497e9c	Ransom.Rapid	Ransom:Win32/Robinhood.AR!MTB
7faeb64c50cd15d036ca259a047d6c62ed491fff3729433fefba0b02c059d5ed	Ransom.Ryuk	Ransom:Win32/Sodinokibi.AD!MTB
81689f1be92c8fb7e94fe241441c7eb43cfb77c6d23592b0248566bd709ff2ed	---	Ransom:Win32/Revil.SI!MTB
d6b7b27e13700aaa7f108bf9e76473717a7a1665198e9aafcc2d2227ca11bba9	Ransom.Ryuk	Ransom:Win32/Ruyk.A!ibt
e8a091a84dd2ea7ee429135ff48e9f48f7787637ccb79f6c3eb42f34588bc684	Malware.AI.4262356839	Ransom:Win32/Maze
ecfb5c95d0f3d112650ef4047936e8fa5244c21c921f6c7a6963e92abab4949d	Ransom.FileCryptor	Ransom:MSIL/Kelnoc.A
f2c9ae3735430b930a81148c0bb470fcb733e456a2a942f859a1b59c4a7b2150	Malware.AI.2057448532	Ransom:Win32/Locky.A
fd32cec288cec4f16dc5430cf86dc17e1d4cf941d635979fc17a59c8d6d83d44	Ransom.PovIsomware	Ransom:MSIL/RnToad.SL!MTB

METHODOLOGY USED FOR THE SYSTEM:

The NAS-Based Ransomware Detection and Monitoring System is designed to provide a robust, intelligent, and automated solution for securing stored data in network-attached storage environments. The system combines machine learning algorithms, real-time file behavior monitoring, IoT alert mechanisms, and threat intelligence integration to detect and mitigate ransomware attacks effectively.

The overall workflow begins with continuous file and system monitoring, followed by anomaly detection using the Isolation Forest model, real-time user alerts, file quarantine, and verification with AlienVault OTX for external threat intelligence. The entire system operates in a local network setup, ensuring real-time communication between the NAS, server, and IoT modules.

i) System Architecture Overview:

The architecture of the NAS ransomware detection system is divided into three primary layers, each with specific responsibilities:

1. Monitoring and Detection Layer:
   This layer, implemented through theNAS monitor module , continuously monitors NAS activities and extracts behavioral features that describe file system operations. It forms the input to the anomaly detection algorithm.

2. File Behavior Analysis Layer:
   This layer uses the file monitor module to identify and record low-level file activities such as creation, modification, deletion, and encryption. It also calculates file entropy and hash values to identify possible encryption attempts.
3. Alert and Response Layer:
   The ESP32-based IoT interface is responsible for alerting users through physical indicators (buzzer, LED) when ransomware-like activity is detected.And the alert also sent to the dashboard and through email for the better security. This ensures immediate awareness even when users are not monitoring the dashboard.

All these layers communicate through a Flask-based server, where data is visualized in a real-time dashboard showing alerts, quarantined files, logs, and system health status.

ii) Data Monitoring and Feature Extraction

At the core of the detection process is the data monitoring phase, where various system and file metrics are continuously collected. The NAS monitoring module tracks these key indicators that collectively represent the behavior of the storage environment.
The extracted features include:

- File Creations: Counts the number of new files added to the NAS within a specific time window. Sudden spikes may indicate automated ransomware-generated files.
- File Deletions: Tracks deleted files, as ransomware often deletes originals after encryption.
- File Modifications: Detects frequent modifications that could signal ongoing encryption or data corruption.
- Suspicious Extensions: Identifies potentially harmful file types (e.g., .exe, .crypt, .locky) often used by ransomware.
- Average Write Size: Observes the average size of written data; unusual write patterns may suggest bulk encryption.
- Entropy Score: Calculates the randomness level in file contents; a sharp rise in entropy indicates potential file encryption.
- SD I/O Rate: Monitors disk input/output activity to detect high-frequency read/write operations typical of ransomware.
- Memory Usage: Tracks RAM consumption, which often spikes during large-scale encryption events.
- File Encryptions: Records files identified as encrypted based on hash and entropy analysis.

These metrics are continuously streamed to the anomaly detection model for analysis. The goal is to build a dynamic behavioral profile of the NAS system, enabling the model to distinguish between normal and suspicious operations in real time.

iii) **Anomaly Detection using Isolation Forest**

To detect ransomware efficiently, the system employs the Isolation Forest algorithm, a machine learning-based anomaly detection model known for its ability to isolate rare or abnormal patterns. Unlike supervised models that require labeled datasets, Isolation Forest is unsupervised, making it suitable for real-time environments where ransomware behaviors evolve rapidly.

Working Principle:

Isolation Forest works by randomly selecting a feature and then choosing a random split value between the minimum and maximum values of that feature. It recursively partitions the data, and anomalies are isolated quickly because they differ significantly from normal data points.

In this system, the model is trained on normal file operation metrics collected during a stable NAS state. When new data arrives, the model evaluates it and assigns an anomaly score. If the score exceeds a predefined threshold, the system flags it as a potential ransomware event.

Advantages of Isolation Forest:

- Works effectively with multi-dimensional behavioral data.
- Detects zero-day ransomware attacks due to pattern-based detection.
- Requires minimal computational resources compared to deep learning models.
- Provides explainable results for audit and reporting.

Once an anomaly is detected, the system transitions to the next stage — file isolation and alert generation.

iv) File Behavior Monitoring and Encryption Detection

The file_monitor.py module handles real-time file-level surveillance. It uses a monitoring algorithm to detect file system events such as creation, renaming, modification, and deletion in the Network Attached Storage. It basically downloads all the files in the NAS and identifies it's hash, entropy,

For encryption detection, the system employs two complementary techniques:

1. Hash Comparison:
   Each file's cryptographic hash (using SHA-256) is computed before and after modification. A change in hash, without a legitimate user operation, signals tampering or encryption.
2. Entropy Analysis:
   The entropy of each file is calculated to determine its randomness level.
   - Normal text or image files have low to moderate entropy.
   - Encrypted or compressed files exhibit high entropy (close to 8 bits per byte).

A sudden increase in entropy combined with a hash mismatch strongly suggests that the file has been encrypted. In such cases, the file is automatically quarantined to prevent further infection. And also identifies if any file operations are detected.

v) Quarantine and Incident Response

When the Isolation Forest model or file monitoring system identifies ransomware-like behavior, the affected files are immediately moved to a quarantine directory within the NAS. This directory is access-restricted and disconnected from the main storage to prevent lateral spread of the ransomware.

Each quarantined file is tagged with:

- File name and location
- Timestamp of detection
- Reason for quarantine (e.g., high entropy, hash mismatch, suspicious extension)

This data is simultaneously logged in the system's database and displayed on the Flask dashboard for user review. Administrators can either permanently delete, restore, or submit quarantined files for deeper analysis.

vi) Alert and Notification Mechanism

To ensure immediate user awareness, the system generates alerts through multiple communication channels:

1. Web Dashboard Alerts:

The Flask-based dashboard provides real-time visual feedback. It contains:

- Alerts Panel: Lists current and past detections with reasons.
- Quarantine Panel: Displays isolated files and their status.
- Activity Log: Summarizes system events, anomalies, and mitigation steps.

2. Email Notifications:

Whenever an anomaly is detected, an automatic email is sent to the registered user or administrator containing:

- File details
- Anomaly detection result with the score

This ensures that even if the administrator is offline, they remain informed about security events.

vi) Integration with Threat Intelligence (AlienVault OTX)

To enhance the accuracy and reliability of detection, the system integrates with AlienVault Open Threat Exchange (OTX) — an open-source, crowd-sourced platform that provides real-time threat intelligence.

When a suspicious file, IP address, or hash is identified:

1. The system queries AlienVault OTX through its API.
2. The OTX database is checked for known indicators of compromise (IOCs).
3. If a match is found, the dashboard updates the file's status to "Confirmed Threat" and includes OTX's additional context (such as malware family, severity, and related campaigns).

This integration ensures that the system is continuously updated with the latest global ransomware trends and can detect both known and emerging variants.

And for every hour the threat intel will get updated that enables the system to be real-time monitoring of the ransomeware.

vi) Continuous Learning and Model Adaptation

The NAS-based ransomware detection system is built to learn and adapt. Over time, the system collects data on both normal and abnormal operations, which are used to retrain the Isolation Forest model.

This retraining helps the model:

- Adjust to the evolving file access patterns.
- Reduce false positives.
- Improve accuracy in distinguishing between legitimate user activity and actual ransomware attacks.

Advantages of the Methodology used for the system:

- Real-time Monitoring: Continuous observation of file behavior ensures early detection.
- Automated Anomaly Detection: Isolation Forest identifies unusual activities without manual intervention.
- IoT Integration: Physical alerts through ESP32 enhance real-world responsiveness. And the integration of ml model for the detection of ransomeware in NAS enhances the monitoring of the system
- Threat Intelligence Validation: AlienVault OTX confirms threats using global datasets.
- Automatic Quarantine: Prevents ransomware spread without affecting unaffected data.
- User Transparency: Dashboard and email reports keep users informed at all times.
- Adaptive Learning: The system improves accuracy through continuous retraining.

LITERATURE SURVEY:

TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
A Systematic Literature Review of Ransomware Detection Methods and Tools for Mitigating Potential Attack [1].	Review of signature-based, heuristic, behavioral, and machine learning approaches .	Provides a broad overview, outlines strengths and weaknesses, identifies research gaps.	Lacks experimental data, mostly theoretical.
Ransomware Detection Using the Dynamic Analysis and Machine Learning [2] .	Uses ML algorithms such as SVM, Random Forest, Decision Trees for behavior-based detection.	Accurate detection, adaptable to new ransomware variants.	Needs large datasets, potential false positives.
Dynamic Feature Dataset for Ransomware Detection Using Machine Learning Algorithms [3].	Dynamic analysis with cuckoo sandbox, dataset creation across Windows platforms .	High accuracy, early detection, applicable to multiple platforms.	Long initial processing times, lower performance on some models.
A Machine Learning-Based Ransomware Detection Method[5].	Data acquisition. feature extraction (entropy,size, type); pre-processing; dataset configuration.	High precision. detects FPE-encrypted files .	Computational complexity; decreased performance in extreme ratios.
Mitigating Storage Challenges through Configuring NAS Using Raspberry Pi[4].	Building low-cost NAS using Raspberry Pi hardware, Samba server .	Cost-effective, customizable, avoids cloud subscription fees.	Limited by Raspberry Pi hardware specs, network-dependent speeds.
Storage Network Attached Using Raspberry Pi [6] .	NAS with Open Media Vault OS,file sharing and remote access.	Energy efficient; low cost; suitable for small offices .	Limited processing power, potential bottlenecks in network .
Network Traffic based Ransomeware Detection [7].	Analyzes network traffic and behavior; uses ransomware signatures database .	Real-time detection, enhances the storage security.	More expensive, reliance on signatures based approach.
Enhancing Healthcare Data Security: Mitigating Ransomware Threats to NAS [10].	Behavioral analytics combined with NAS in healthcare with compliance monitoring .	Reduces downtime, improves compliance, enhances patient care.	Specialized domain, complexity in setup.

Honeypot-based Ransomware Detection as a Component of Security Posture Monitoring in Zero Trust Architecture [9].	Setup of decoy files to attract ransomware and analyze behavior.	Early ransomware identification without impact on real data .	May not detect all ransomware, limited by attack targeting .
Detection of Ransomware Attacks Using Processor and Disk Usage Data [12].	Uses CPU and disk usage anomaly detection .	Non-intrusive, detects unusual system behavior.	Variable resource usage may cause false positives.
Encrypted NAS using Raspberry Pi 4 [15].	Use Raspberry Pi 4 with OpenMediaVault, encrypt files on NAS and control access via Samba .	Strong encryption and integrity; low power; cost-effective .	Overhead on encryption/decryption; limited scalability.
Cryptographic Ransomware Encryption Detection: Survey [16].	Surveys focus on cryptographic ransomware detection techniques	Covers state-of-art crypto ransomware detection .	Focused mainly on cryptographic methods not on others.
Ransomware Detection with Machine Learning: A Systematic Review [17].	Systematic review of ML-based ransomware detection .	Highlights effective ML techniques and features	Some methods less explored in practice
Exploring Ransomware Detection Based on AI [20].	Uses AI techniques for ransomware detection.	Discusses latest AI-powered detection techniques .	Practical deployment challenges
Ransomware Detection using Memory Analysis [23].	Uses memory and runtime analysis for detection .	Effective for in-memory ransomware.	May miss early-stage behaviors
XRan: Explainable Deep Learning-Based Detection [24].	Deep Learning with explainability for ransomware detection.	High accuracy with interpretable results .	Requires extensive computational resources.

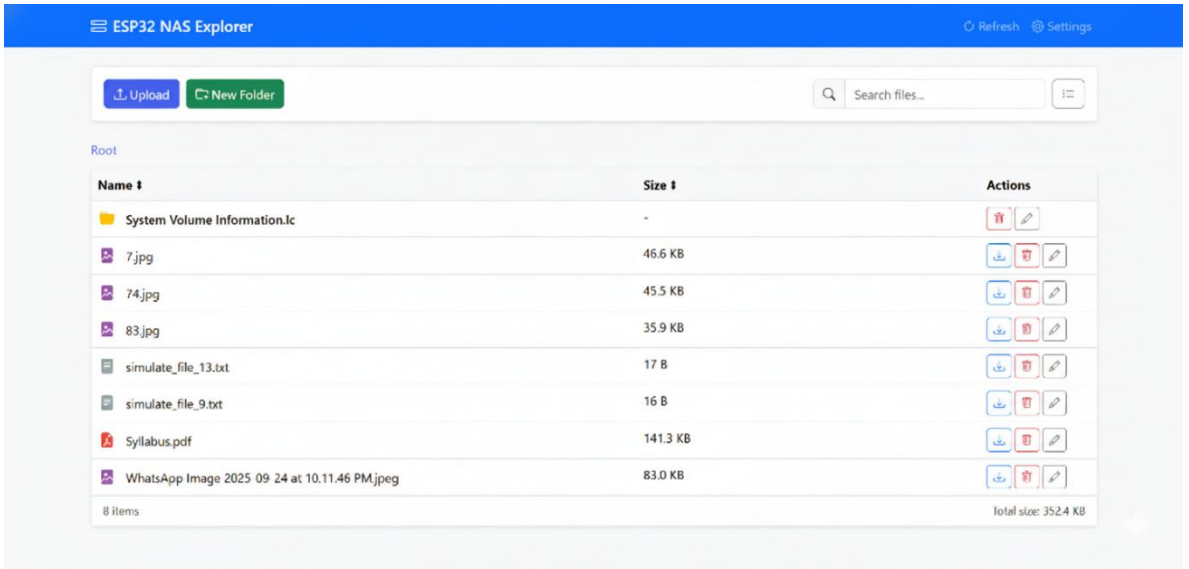
IMPLEMENTATION:

i) Software

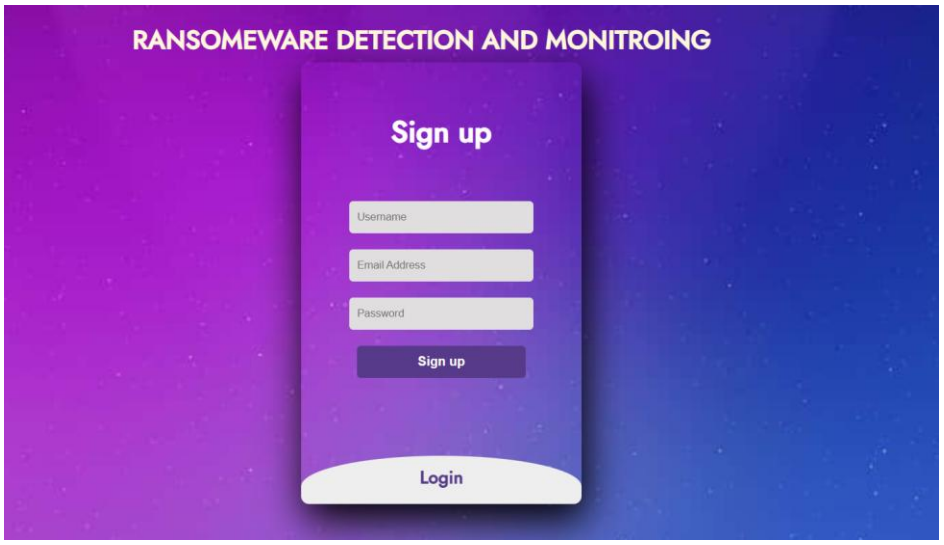
The entire system runs through the flask via the api request from the endpoints. Once the monitoring loops starts it will continuously monitor and detects the ransomeware in the Network attached storage.

NAS EXPLORER:

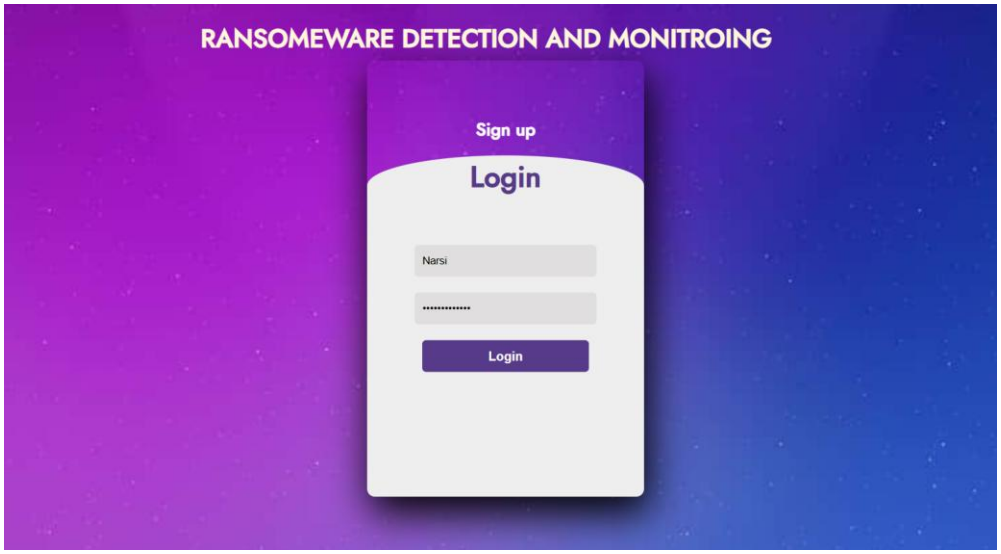
To check if all the operations on the NAS are working properly and also to perform attacks and to determine if the ransomeware detection works properly



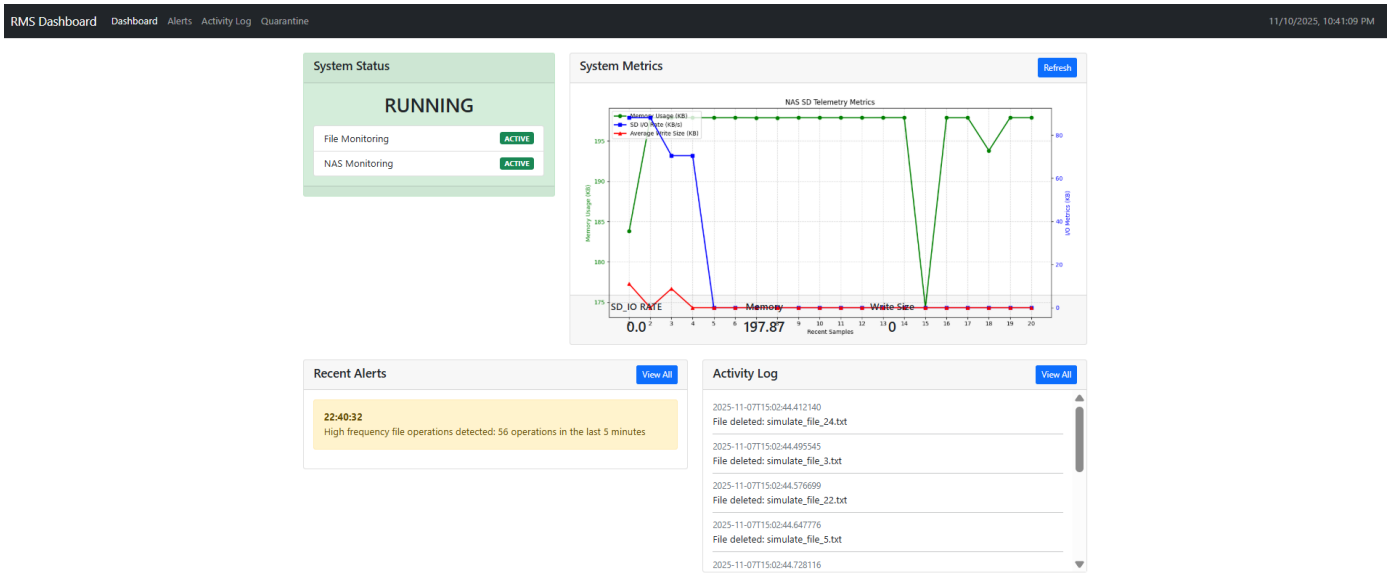
SIGNUP;



LOGIN:



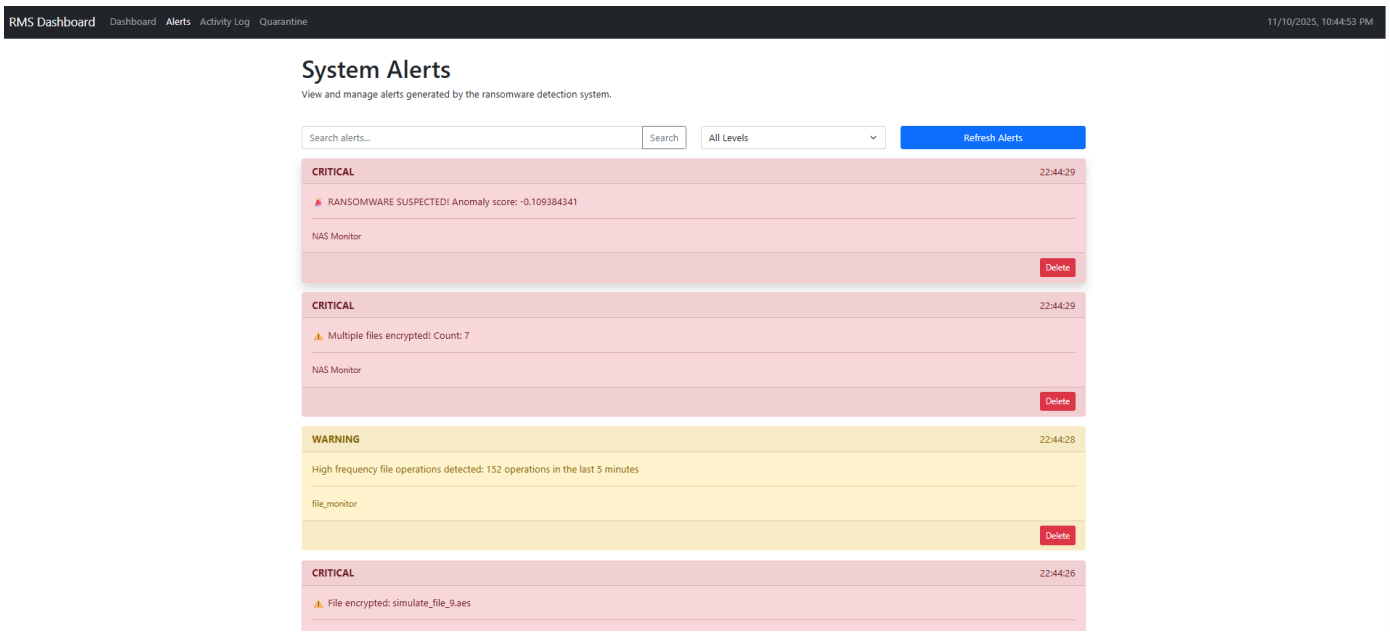
RMS DASHBOARD:



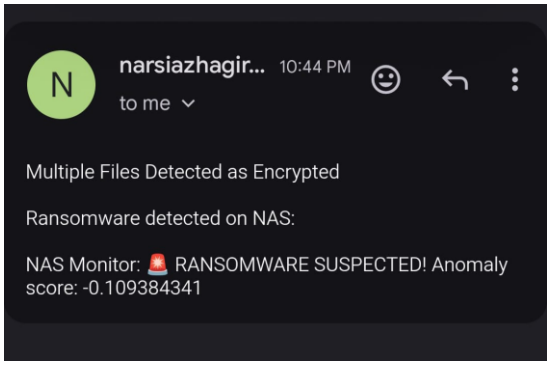
ALERTS:

Once the hacker attacks the NAS this system will monitor continuously and identify the anomaly and send alerts to user via dashboard and also through the mail.

Dashboard alerts :

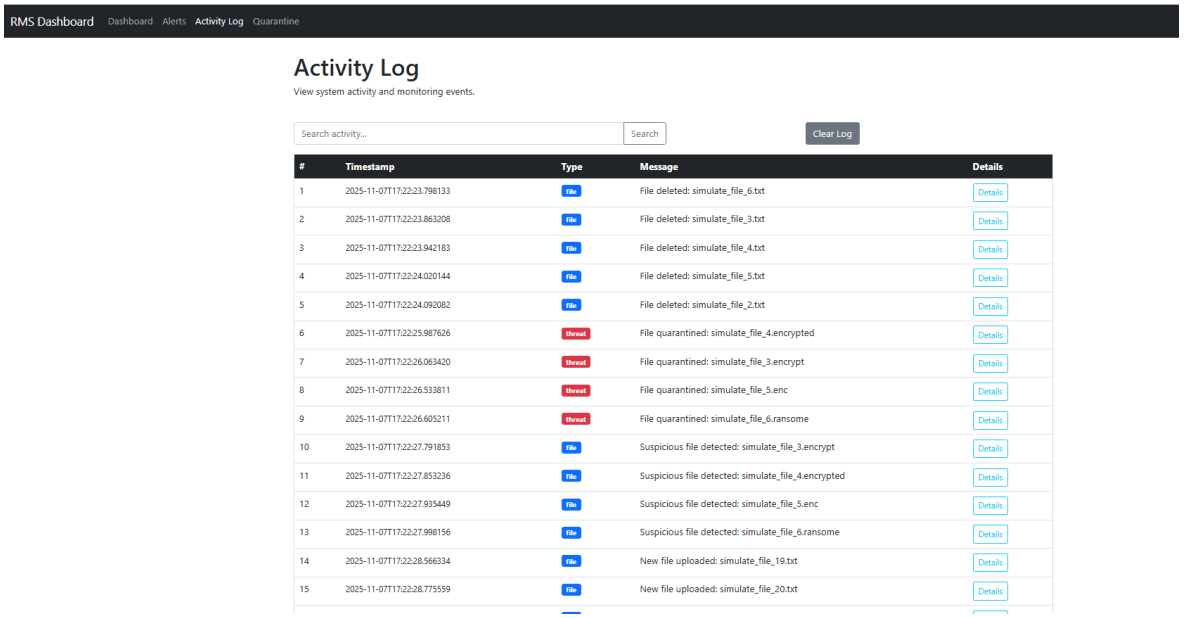


E-mail alerts:



ACTIVITY LOGS:

To check what are the operations has been carried out in Network Attached Storage.



#	Timestamp	Type	Message	Details
1	2025-11-07T17:22:23.798133	file	File deleted: simulate_file_6.txt	Details
2	2025-11-07T17:22:23.863208	file	File deleted: simulate_file_3.txt	Details
3	2025-11-07T17:22:23.942183	file	File deleted: simulate_file_4.txt	Details
4	2025-11-07T17:22:24.020144	file	File deleted: simulate_file_5.txt	Details
5	2025-11-07T17:22:24.092082	file	File deleted: simulate_file_2.txt	Details
6	2025-11-07T17:22:25.987626	threat	File quarantined: simulate_file_4.encrypted	Details
7	2025-11-07T17:22:26.063420	threat	File quarantined: simulate_file_3.encrypt	Details
8	2025-11-07T17:22:26.533811	threat	File quarantined: simulate_file_5.enc	Details
9	2025-11-07T17:22:26.605211	threat	File quarantined: simulate_file_6.ransome	Details
10	2025-11-07T17:22:27.791853	file	Suspicious file detected: simulate_file_3.encrypt	Details
11	2025-11-07T17:22:27.853236	file	Suspicious file detected: simulate_file_4.encrypted	Details
12	2025-11-07T17:22:27.935449	file	Suspicious file detected: simulate_file_5.enc	Details
13	2025-11-07T17:22:27.998156	file	Suspicious file detected: simulate_file_6.ransome	Details
14	2025-11-07T17:22:28.566334	file	New file uploaded: simulate_file_19.txt	Details
15	2025-11-07T17:22:28.775559	file	New file uploaded: simulate_file_20.txt	Details

39	2025-11-07T17:24:26.601136	system	NAS ransomware detection system stopped	
40	2025-11-10T22:38:44.358974	system	NAS monitoring started	
41	2025-11-10T22:38:48.037624	system	NAS ransomware detection system started	
42	2025-11-10T22:38:48.084890	system	NAS Monitor started for 10.167.93.24:8080	
43	2025-11-10T22:38:48.129262	system	File Monitor started for 10.167.93.24:8080	
44	2025-11-10T22:38:48.130671	system	NAS monitoring started	
45	2025-11-10T22:43:23.742496	file	New file uploaded: simulate_file_0.txt	Details
46	2025-11-10T22:43:24.297665	file	New file uploaded: simulate_file_1.txt	Details
47	2025-11-10T22:44:27.088905	file	New file uploaded: simulate_file_2.txt	Details
48	2025-11-10T22:44:27.131829	file	New file uploaded: simulate_file_3.txt	Details
49	2025-11-10T22:44:27.213807	file	New file uploaded: simulate_file_4.txt	Details
50	2025-11-10T22:44:27.272426	file	New file uploaded: simulate_file_5.txt	Details
51	2025-11-10T22:44:27.348882	file	New file uploaded: simulate_file_6.txt	Details
52	2025-11-10T22:44:27.410937	file	New file uploaded: simulate_file_7.txt	Details
53	2025-11-10T22:44:27.492781	file	New file uploaded: simulate_file_8.txt	Details
54	2025-11-10T22:44:27.582314	file	New file uploaded: simulate_file_10.txt	Details
55	2025-11-10T22:44:27.640894	file	New file uploaded: simulate_file_11.txt	Details
56	2025-11-10T22:44:27.701625	file	New file uploaded: simulate_file_12.txt	Details
57	2025-11-10T22:44:27.756412	file	New file uploaded: simulate_file_14.txt	Details
58	2025-11-10T22:44:27.817448	file	New file uploaded: simulate_file_15.txt	Details
59	2025-11-10T22:44:27.876036	file	New file uploaded: simulate_file_16.txt	Details
60	2025-11-10T22:44:28.036488	file	New file uploaded: simulate_file_17.txt	Details

QUARANTINE:

If any suspicious file is detected it will automatically quarantine those files and remove those files from NAS.

RMS Dashboard Dashboard Alerts Activity Log Quarantine

Quarantine

Manage quarantined files and restore them to NAS.

Quarantine a File

Filename

example.exe

Reason

Suspicious behavior

Quarantine File

Action Result

Results of mitigation actions will appear here.

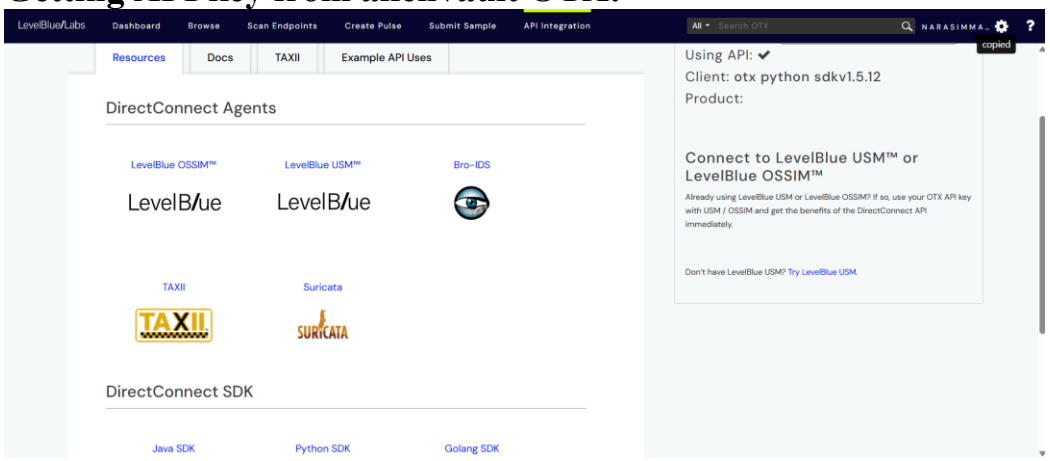
Quarantined Files

Filename	Quarantine Time	Reason	Actions
simulate_file_0.locked	2025-11-10T22:45:03.679904	Suspicious Behaviour Detected	Restore Delete
simulate_file_1.crypt	2025-11-10T22:44:50.775737	Suspicious Behaviour Detected	Restore Delete
simulate_file_2.aes	2025-11-10T22:44:50.972756	Suspicious Behaviour Detected	Restore Delete
simulate_file_3.encrypt	2025-11-10T22:46:11.057932	Suspicious Behaviour Detected	Restore Delete
simulate_file_4.encrypted	2025-11-10T22:45:52.013968	Suspicious Behaviour Detected	Restore Delete
simulate_file_5.enc	2025-11-10T22:45:01.937319	Suspicious Behaviour Detected	Restore Delete
simulate_file_6.ransome	2025-11-10T22:45:02.159532	Suspicious Behaviour Detected	Restore Delete
simulate_file_7.locked	2025-11-10T22:44:44.146707	Suspicious Behaviour Detected	Restore Delete
simulate_file_8.crypt	2025-11-10T22:44:49.527870	Suspicious Behaviour Detected	Restore Delete

INTEGRATION WITH ALIENVAULT OTX:

It is an open source threat intel exchange platform where this system will be fetching the real time threat intel using the api key and compares if any of the real-time threats exist in our system.

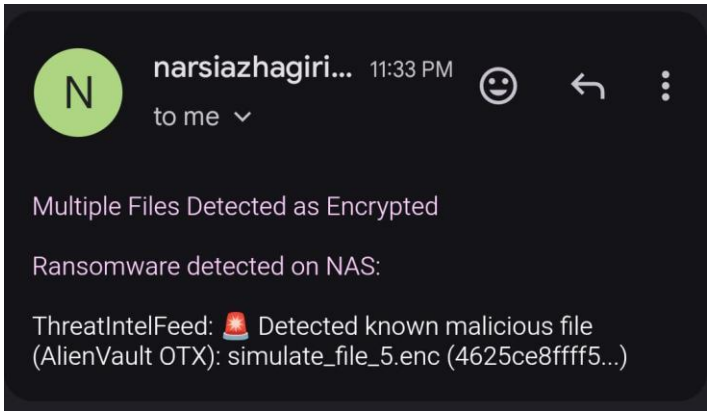
Getting API key from alienvault OTX:



Fetching threat intels from through the API integration:

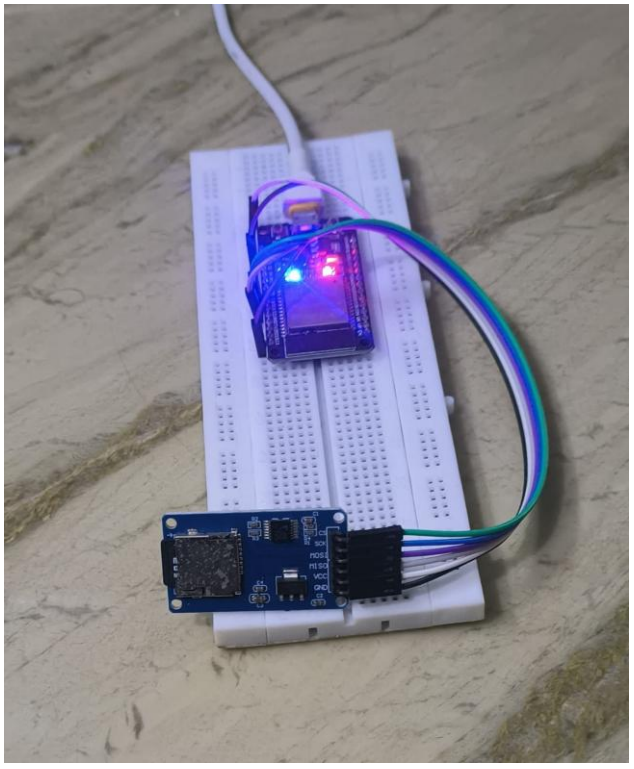

```
"data": {
  "hashes": [
    "4625ce8ffff58c7897237d97f6c8485544c9b485611790b0303b2de5b441e27e",
    "6e7244523e8b853c4bc3dd655f5dd40b59fee0b9e1fa70c8bf829dee315d731e",
    "fb4a36fbdcf51c9847d94c0fcf8d62045a5ab781b7b1695eaed5b22f21d34ed9",
    "0cccfcc2a5813d2617213d2c24fe96edd360ff23d25d4f6c5b7c2ad6f3f3e87e",
    "338800bd013fc0bc8bad5e6cfaa723c6a0d35ebbe8c2ba05ca10c03b0886d8a3",
    "a770962691edc713ffbba74b855ae59bc32735a7e6524282a5a1f813e971a313",
    "10c873fe71ec276259331ec99d52e713843c8a2c035855c8b7578210001c55bf",
    "2a79aff27e084de07a0d20704c2115fcfa78fd0204aa3a07c83c4eb64ff3e8f2",
    "776df17dcfb99b5cc06d1ff75528955ca51b5f6d115ec4e6ff0151fff4190c30",
    "b011de113a0ba863bb987f21b75b1ffa835f69c02a856a5b91c61b6a8549da2e",
    "c640b4d8201c8a738d50f7cc8f842a4875ff24c5440f152cecce4af99f3ddbdd"
  ],
  "ips": [
    "45.8.146.93",
    "104.247.162.67",
    "66.29.144.75",
    "193.39.187.165",
    "103.35.190.170",
    "45.159.248.110",
    "116.125.126.38",
    "86.104.72.247",
    "103.231.75.101"
  ],
  "domains": [
    "galaterrace.com",
    "pierregems.com",
    "unlockk.play-google-store.ru",
    "goolgepalay.com",
    "www.scoringmnmathleague.org",
    "tikpremplaymarktuzb.sbs",
    "tiktokgplya.sbs",
    "www.tikitok-playgoolge.sbs",
```

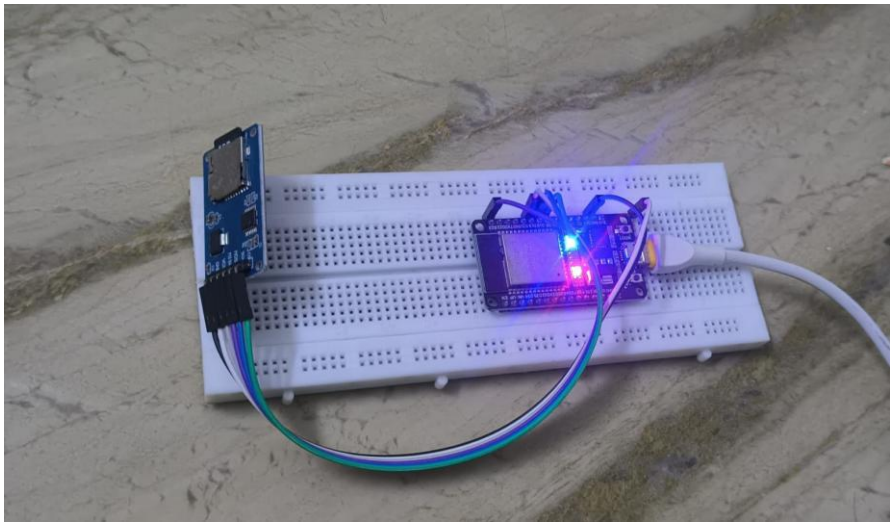

Alerts :



ii) Hardware

In the hardware implementation the NAS acts as a main role to monitor and detect the anomaly operations in that and an esp32 microcontroller is integrated using wifi and both hardware and software should be connected with same wifi in order to perform the ransomeware detection and monitoring. In which the system uses the trained isolation forest model to make the anomaly detection so that it sends alert to the user and the affected files will be automatically removed from the Network Attached Storage.





CONCLUSION:

The NAS-Based Ransomware Detection System provides improved protection for network-attached storage environments. By continuously monitoring in real time and using anomaly detection techniques, this system can accurately identify ransomware activities. It is designed specifically for NAS platforms, ensuring smooth integration with NAS protocols and hardware while keeping overall performance effect to a minimum. It uses smart anomaly detection techniques to spot these threats quickly and accurately. Additionally, the system sends instant alerts to notify the relevant people whenever a possible ransomware attack is detected, so you can take right away action.

This system's key strengths include proactive threat detection, automated actions like isolating infected files or blocking harmful IP addresses, logging of suspicious events for auditing and forensic purposes, and immediate alert notifications. This approach reduces data loss, minimizes downtime, and improves overall NAS security.

Future Enhancements:

To further strengthen the system, several future enhancements can be considered:

- Improved Pre-Encryption Detection: Creating machine learning algorithms designed to identify ransomware prior to the initiation of encryption. This can lower potential harm and facilitate proactive defenses [15].
- Multi-Layered and Hybrid Detection: Combining several machine learning models to monitor and identify ransomware with higher accuracy.
- Better Datasets and Real-world Testing: Developing more representative datasets tailored for NAS-based scenarios and executing real-world evaluations [3].
- Integration with Automated Response: Seamlessly linking NAS detection modules with automated quarantine, alerting, and backup or recovery methods to provide whole protection.

Implementing these enhancements can transform the NAS-Based Ransomware Detection System into a more intelligent, holistic, and user-friendly security measure. It will adapt to evolving cyber threats while offering robust protection for critical data in NAS environments for years ahead.
As cyber threats grow more intricate, such solutions will be crucial for safeguarding data, maintaining business continuity, and preserving trust in digital storage systems. The efforts made here extend beyond mere file protection; they are about ensuring organizations and individuals that their most vital information will remain safe, secure, and readily available whenever needed.

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